

Fig. 4: Actual-size PCB of the basic circuit with a relay

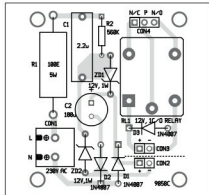


Fig. 5: Component layout of the PCB in Fig. 4

interfaced to grid power. There is no transformer for power-line isolation in the circuit, so the user must be very careful with the circuit. An isolation transformer (1:1) should be used when probing the assembled circuit.

If you want a suitable relay driver circuit with galvanic isolation, so that your low-power circuitry can control the current through the relay safely, try the add-on circuit shown in Fig. 3. It is built around optocoupler MCT2E (IC1), timer LM555 (IC2) and a few discrete components. The op-

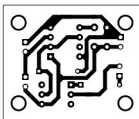


Fig. 6: Actual-size PCB of the add-on circuit

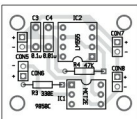


Fig. 7: Component layout of the PCB in Fig. 6

PARTS LIST

Semiconductors:

- IC1 - MCT2E optocoupler
- IC2 - LM555 timer
- D1-D3 - 1N4007 rectifier diode
- ZD1, ZD2 - 12V, 1W zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 100-ohm, 5W
- R2 - 560-kilo-ohm
- R3 - 330-ohm
- R4 - 47-kilo-ohm

Capacitors:

- C1 - 2.2µF, 400V polyester
- C2 - 100µF, 35V electrolytic
- C3 - 0.1µF ceramic disc
- C4 - 0.01µF ceramic disc

Miscellaneous:

- RL1 - 12V, 1 C/O relay
- CON1 - 2-pin connector terminal
- CON2, CON3, CON5-CON8 - 2-pin connector
- CON4 - 3-pin connector

on the input voltage and the optocoupler being used.

While using the add-on circuit, disconnect CON2 from CON3 to remove the relay circuit with diode D3 from the basic circuit. Connect CON3 to either

CON7 or CON8 in the add-on circuit.

DC supply available across CON2 may be used by connecting it to CON5. The add-on circuit operates when you provide high voltage of, say, 5V across CON6. The relay connected across CON8 energises.

On the other hand, when there is no voltage across CON6, the relay connected across CON7 energises. You can enclose both the circuits (basic circuit and add-on circuit) in a suitable plastic case/prototype enclosure.

Construction and testing

An actual-size, single-side PCB for the basic circuit is shown in Fig. 4 and its component layout in Fig. 5.

Similarly, an actual-size, single-side PCB for the add-on circuit is shown in Fig. 6 and its component layout in Fig. 7.

Enclose both the circuits in a suitable small box in such a way that connectors CON1 through CON8 can be easily accessed, as and when required. Ensure proper wiring to avoid any mistake. ●



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